DBMS -- 3-Tier Architecture

Target: Explain 3-tier architecture in 30s + name each layer's job clearly.

What Is 3-Tier Architecture?
A way of organizing a web application into 3 separate layers, each with a single responsibility.
All web applications are essentially 3-tier in nature.

```
        USER (Browser / Mobile App)
              |
              | HTTP Request
    PRESENTATION LAYER      Client Layer (UI)
    (Client 1, 2, 3)

          |
          | Business logic call

   APPLICATION LAYER       Business Layer (App Server)
   (Application Server
   / Business Server)

          |
          | DB query

     DATA LAYER          Database Layer
    (Database Server)
```

The 3 Layers -- What Each Does

Layer	Also Called	Responsibility	Examples
Presentation Layer	Client Layer	What the user sees and interacts with	Browser, Mobile App, React UI
Application Layer	Business Layer	Processes requests, applies business logic, acts as middleman	Node.js, Express, Spring Boot
Data Layer	Database Layer	Stores and retrieves actual data	MySQL, PostgreSQL, MongoDB

Key Rules from the Ebook

No direct communication between Client and Database -- the Application layer always sits in between
Application Server = the middle layer; exchanges partially processed data between client and DB
Schema = a logical representation of the database. One database can have multiple schemas.

Why 3-Tier? (Interview Explanation)

Problem with 1-tier / 2-tier
How 3-tier solves it

Client talks directly to DB -- security risk
App layer acts as a gatekeeper

Business logic mixed with UI -- hard to maintain
Separation of concerns

Scaling is hard if everything is together
Each layer can scale independently

One change breaks everything
Changes in one layer don't affect others

Real-World Example (Your Work at Berkadia)
```
Browser (React UI)
Node.js / Express API Server     middle layer, applies auth (Keycloak), business logic
PostgreSQL / Elasticsearch       stores documents, search indexes
```

Your 30-Second Script

"3-tier architecture splits an application into three layers. The presentation layer is what the user sees -- the browser or mobile app. The application layer, also called the business layer, sits in the middle -- it processes requests, applies business logic, and talks to both sides. The data layer is the database -- it stores and retrieves data. The key rule is that the client never talks directly to the database -- all communication goes through the application server."

Follow-Up Questions (Expect These)
Q: What is the difference between 2-tier and 3-tier architecture?

In 2-tier, the client talks directly to the database -- no application server. Faster but less secure and harder to scale. 3-tier adds the middle application layer for security, logic separation, and scalability.

Q: What is a schema in DBMS?

A schema is a logical representation of the database -- it defines the tables, columns, relationships, and constraints. One database server can host multiple schemas.

Q: Can there be more than 3 tiers?

Yes -- n-tier architecture. Microservices is an example where the application layer is split into many independent services. But the classic interview answer is 3-tier.